

A Projector-based Full Windshield HUD Simulator to Evaluate the Visualization Methods

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Abstract— Many researchers in automotive industry are developing various methods for warning the dangerous situations to the driver. In order to evaluate the visualization methods of the dangerous vehicles in the projector-based full windshield head-up display (FW-HUD) system, we develop a visualization simulator of FW-HUD. The simulator is implemented by two-types of projectors and the homography based calibration between the windshield and the wall. The FW-HUD simulator is made like the real vehicle to feel the sense of reality.

Keywords— FW-HUD, Vehicle visualization, information visualization, implement simulator.

I. INTRODUCTION

Today, many luxury vehicles are equipped with Head-Up Display (HUD) system to effectively visualize various information while driving [1]. Especially, many automobile companies have developed HUD system with small display, which is not immersive than full windshield head-up display (FW-HUD) system using the windshield display. Thus, they have developed augmented reality based FW-HUD system to intuitively visualize the augmented safety information such as the location of detected vehicle and pedestrian [2]. However, the intuitive visualization using FW-HUD system is evaluated by the repetitive experiments under same environment. Therefore, we suggest a projector-based FW-HUD simulator as indoor test-bed.

During recent decades, various full windshield head up display systems are developed by many researchers. Akihiko Sato et al. [3] suggest windshield head-up display based visual navigation system. In order to augment the navigation information, they made Windshield HUD system. The system is implemented by reflection of the projection image using the projector which is established on the vehicle's roof. The projection image is reflected to the windshield using a mirror. Wen Wu et al. [4] made the FW-HUD system to augment the landmark, which is the line drawn by a laser projector. Sun Innovations co. [5] developed the FW-HUD Development Kit. It includes upgraded miniature laser projector that supports the dual colors. But, it can visualize the restricted expression and is also expensive.

The proposed simulator can effectively visualize the various information for virtual driving. The visualization prevents the distracted driving by warning the dangerous vehicles while driving. Therefore, we inexpensively implement an indoor simulator to experiment the visualization for warning of dangerous obstacles.

II. PROJECTOR-BASED FW-HUD SIMULATOR

In this section, we describe the system setup and calibration of projector-based FW-HUD simulator. The system is made using several devices of low price and simple calibration.

A. System Setup

The simulator is made to be similar to real vehicle in order to feel like a vehicle. Figure 1 illustrates the concept of projector-based full windshield HUD simulator. The simulator means the indoor system to visualize the road scene and information of detected vehicles.

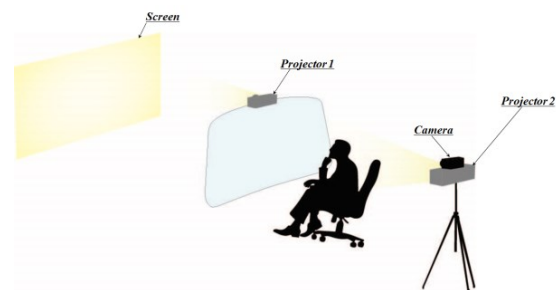


Figure 1. Concept of projector-based FW-HUD simulator.

TABLE I. SPECIFICATION OF SYSTEM COMPONENTS

Type	Name	Specification	Company
Reflection Film	VIVI-WR (Clear)	1.6m x 1.5m	ViViTech
DLP Projector	X2010	3000 ANSI	Optoma
	EX610ST	3000 ANSI	Optoma
Camera	Quick Cam Pro 9000	2M Pixel	Logitech
Windshield	Kia Carens OEM	Laminated Glass	Hapdong

In Figure 1, *Projector1* projects the road scene on the screen and *Projector2* is the device to augment the some information on the windshield. Also, *Camera* is used to track

driver's view position. The curved surface is the windshield and the yellow rectangle means a project screen.

In order to build the simulator, we use a real windshield, two projectors, a camera, and transparent reflection film. The detail specification of system components is described in Table I. Most of all, *EX610ST* is used wide-angle lens because it has to cover to the whole region of the windshield in limited distance. Finally, Figure 2 shows the hardware configuration of our simulator.



Figure 2. Hardware configuration of projector-based FW-HUD simulator.

B. System Calibration

In order to augment the useful information, the proposed simulator performs the calibration based on fixed driver's head position [3, 4]. In this case, we first suppose that the windshield is a flat plane. Therefore, we apply the homography based calibration method to align between the detection region projected on the windshield and obstacles of road scene. The calibration method performs two stages. In the first stage, we remove the distortion of the image projected to the windshield by *Projector2*. In this case, the proposed system uses the homography between the original image and projected image on the windshield. The unwarped image to remove the distortion is computed by the following equation:

$$I_{uw} = H_{iw}^{-1} * I_{in}, \quad (1)$$

where I_{uw} is the unwarped image, I_{in} is the original input image, and H_{iw}^{-1} means inverse homography between the original image and projected image on the windshield. When the unwarped image is projected on the windshield, the image is displayed without distortion.

After the distortion of projected image on the windshield is removed, we perform the alignment between the points projected on the windshield and the corresponding points projected on the wall screen in the second stage. In this case, we also compute the homography between the projected image on the wall and the projected unwarped image on the windshield. The final aligned image is represented by the following equation:

$$H_{total} = H_{ww} * H_{iw}^{-1}, \quad (2)$$

$$I_{new} = H_{total} * I_{in}, \quad (3)$$

where H_{ww} means the homography between the windshield and the wall. And, I_{new} is the transformed image to be aligned correctly in the road scene. The proposed system can visualize aligned information by projecting I_{new} on the windshield.

III. EXPERIMENTS

In experiments, the system can visualize the detected vehicles with various shapes. According to *Gestalt Laws of Perceptual Organization*, objects grouped together are seen as a whole by *Law of Closure* [6]. Therefore, we designed the shape of square bracket to visualize the detected vehicles. Figure 3 shows the result of augmented vehicles using shape of designed square bracket.



Figure 3. The result of augmented information on the full windshield.

IV. CONCLUSION

In this paper, we implement the projector-based FW-HUD simulator to experiment the visualization for warning of dangerous obstacles as indoor test-bed. The simulator is made using low cost and simple calibration. And we perform the experiment for visualization of detected vehicles using shape of designed square bracket. In the future, we will suggest the various shapes for effective visualization of dangerous vehicles. And the suggested shapes of visualization will be verified by many tests using the proposed simulator.

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